

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	6	8	19

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QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Ans: Given: $Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}$ $X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$ $\epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{bmatrix}$

Let X For Least squares estimate (LSE) of θ_1 and θ_2 , the sum of squares of errors has to be minimum.

i.e., $\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 = Q(\theta_1, \theta_2)$ has to be minimum

$Q(\theta_1, \theta_2)$ is minimum when $\frac{\partial Q}{\partial \theta_1} = 0$ and $\frac{\partial Q}{\partial \theta_2} = 0$

$\Rightarrow Y_1, Y_2, Y_3$ are the observed values.

$$\frac{\partial Q}{\partial \theta_1} = \frac{\partial}{\partial \theta_1} \left[(Y_1 - \theta_1 - \theta_2)^2 + (Y_2 - \theta_1 + 2\theta_2)^2 + (Y_3 - 2\theta_1 + \theta_2)^2 \right] = 0$$

$$2(Y_1 - \theta_1 - \theta_2)(-1) + 2(Y_2 - \theta_1 + 2\theta_2)(-1) + 2(Y_3 - 2\theta_1 + \theta_2)(-2) = 0$$

$$-Y_1 + \theta_1 + \theta_2 - Y_2 + \theta_1 - 2\theta_2 - 2Y_3 + 4\theta_1 - 2\theta_2 = 0$$

$$6\theta_1 - 3\theta_2 = Y_1 + Y_2 + 2Y_3 \quad \text{--- (1)}$$

and $\frac{\partial Q}{\partial \theta_2} = \frac{\partial}{\partial \theta_2} \left[(Y_1 - \theta_1 - \theta_2)^2 + (Y_2 - \theta_1 + 2\theta_2)^2 + (Y_3 - 2\theta_1 + \theta_2)^2 \right] = 0$

$$2(Y_1 - \theta_1 - \theta_2)(-1) + 2(Y_2 - \theta_1 + 2\theta_2)(2) + 2(Y_3 - 2\theta_1 + \theta_2)(1) = 0$$

$$-Y_1 + \theta_1 + \theta_2 + 2Y_2 - 2\theta_1 + 4\theta_2 + Y_3 - 2\theta_1 + \theta_2 = 0.$$

$$-3\theta_1 + 6\theta_2 = Y_1 - 2Y_2 - Y_3 \quad \text{--- (2)}$$

from (1) and (2)

$$\begin{aligned} 6\theta_1 - 3\theta_2 &= Y_1 + Y_2 + 2Y_3 \\ -6\theta_1 + 12\theta_2 &= 2Y_1 - 4Y_2 - 2Y_3 \end{aligned}$$

$$9\theta_2 = 3Y_1 - 3Y_2$$

$$\boxed{\hat{\theta}_2 = \frac{Y_1 - Y_2}{3}}$$

from (1),

$$6\theta_1 = 2\left(\frac{Y_1 - Y_2}{3}\right) + Y_1 + Y_2 + 2Y_3$$

$$6\theta_1 = 2(Y_1 + Y_3)$$

$$\boxed{\hat{\theta}_1 = \frac{Y_1 + Y_3}{3}}$$

$\therefore$ The LSEs of θ_1 and θ_2 are $\left(\frac{Y_1 + Y_3}{3}\right)$ and $\left(\frac{Y_1 - Y_2}{3}\right)$ respectively.

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

Ans:

(a) For the correlation matrix ρ , let's first calculate covariance matrix of

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}, \quad \sigma^2 = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

as they are uniform random variables,

for a uniform RV on $(0, 1)$, Expectation = $E(U_1) = \int_0^1 u \cdot f_U(u) \cdot du$

$\downarrow$
 pdf, $f_U(u) = 1 \quad \forall u \in (0, 1)$

$$= \int_0^1 u \cdot 1 \cdot du = \left. \frac{u^2}{2} \right|_0^1 = \frac{1}{2}.$$

$$\begin{aligned} \text{Var}(U_1) &= E(U_1^2) - [E(U_1)]^2 \\ &= \int_0^1 u^2 \cdot f_U(u) \cdot du - \left(\frac{1}{2}\right)^2 \\ &= \left. \frac{u^3}{3} \right|_0^1 - \frac{1}{4} \end{aligned}$$

$$\text{Var}(U_1) = \frac{1}{12}.$$

$$\Rightarrow X_1 = U_1 \Rightarrow \mu_1 = \frac{1}{2}, \quad \sigma_1^2 = \sigma_{11} = \frac{1}{12}$$

$$X_2 = U_2 \Rightarrow \mu_2 = \frac{1}{2}, \quad \sigma_2^2 = \sigma_{22} = \frac{1}{12}$$

$$X_3 = U_1 + U_2 \Rightarrow \mu_3 = E(U_1) + E(U_2) = 1,$$

$$\begin{aligned} \sigma_3^2 = \sigma_{33} &= \text{Var}(U_1) + \text{Var}(U_2) \\ &= \frac{1}{12} + \frac{1}{12} \end{aligned}$$

$$\sigma_3^2 = \frac{1}{6}$$

($\because U_1$ and U_2 are independent).

And the covariances, $\sigma_{12} = \text{Cov}(U_1, U_2) = 0$

$$\begin{aligned}\sigma_{32} &= \sigma_{23} = \text{Cov}(U_2, U_1 + U_2) = \text{Cov}(U_2, U_1) + \text{Cov}(U_2, U_2) \\ &= 0 + \text{Var}(U_2) \\ &= \frac{1}{12}\end{aligned}$$

$$\begin{aligned}\sigma_{13} &= \sigma_{31} = \text{Cov}(U_1 + U_2, U_1) = \text{Cov}(U_1, U_1) + \text{Cov}(U_2, U_1) \\ &= \text{Var}(U_1) + 0 \\ &= \frac{1}{12}\end{aligned}$$

So the covariance matrix is,

$$\sigma^2 = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 \end{bmatrix} = \begin{bmatrix} \frac{1}{12} & 0 & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} \end{bmatrix}$$

And the correlation matrix is,

$$\rho = \begin{bmatrix} \frac{\sigma_1^2}{\sigma_1 \cdot \sigma_1} & \frac{\sigma_{12}}{\sigma_1 \cdot \sigma_2} & \frac{\sigma_{13}}{\sigma_1 \cdot \sigma_3} \\ \frac{\sigma_{21}}{\sigma_2 \cdot \sigma_1} & \frac{\sigma_2^2}{\sigma_2 \cdot \sigma_2} & \frac{\sigma_{23}}{\sigma_2 \cdot \sigma_3} \\ \frac{\sigma_{31}}{\sigma_3 \cdot \sigma_1} & \frac{\sigma_{32}}{\sigma_3 \cdot \sigma_2} & \frac{\sigma_3^2}{\sigma_3 \cdot \sigma_3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \sqrt{\frac{1}{6}}} \\ 0 & 1 & \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \sqrt{\frac{1}{6}}} \\ \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \sqrt{\frac{1}{6}}} & \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \sqrt{\frac{1}{6}}} & 1 \end{bmatrix}$$

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$$\rho = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

[Continued (b) in Page 8]

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

- (a) (6 points) Cluster the four items using each of the following procedures:
- Complete linkage hierarchical procedure.
 - Average linkage hierarchical procedure.
- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Ans:

(a)

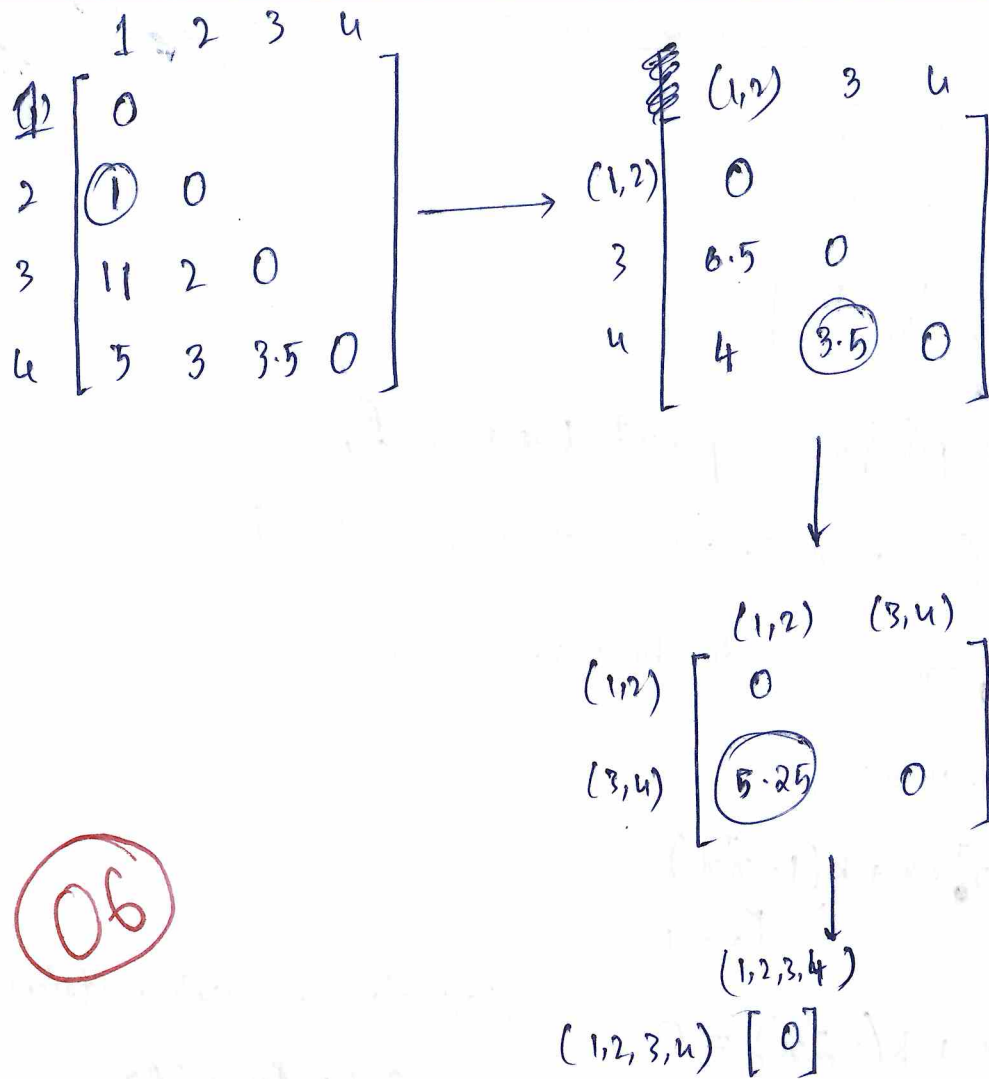
i. Complete Linkage: If the distance between the clusters U and V is minimum, (UV) will be clustered/grouped together.

for others, $\text{dist}(UV, W) = \max(\text{dist}(U, W), \text{dist}(V, W))$.
↙ ↘
earlier distances.

Matrix:

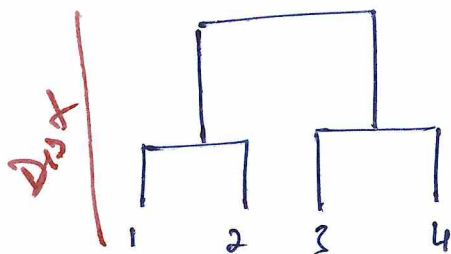
$$\begin{array}{c}
 \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\
 \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix}
 \end{array}
 \longrightarrow
 \begin{array}{c}
 \begin{array}{c} (1,2) \quad 3 \quad 4 \\
 \begin{bmatrix} 0 & & \\ 11 & 0 & \\ 5 & 3.5 & 0 \end{bmatrix}
 \end{array}
 \longrightarrow
 \begin{array}{c}
 \begin{array}{c} (1,2) \quad (3,4) \\
 \begin{bmatrix} 0 & \\ 11 & 0 \end{bmatrix}
 \end{array}
 \end{array}
 \end{array}
 \downarrow
 \begin{array}{c}
 (1,2,3,4) \\
 \begin{bmatrix} 0 \end{bmatrix}
 \end{array}
 \end{array}$$

ii. $\text{dist}(UV, W) = \frac{\text{dist}(U, W) + \text{dist}(V, W)}{2}$

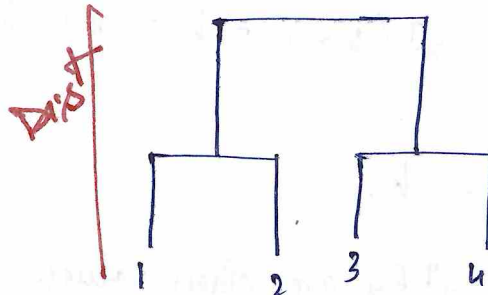


(b) Dendograms :

Complete Linkage



Average Linkage



Q2 ⇒ Both Dendograms are same as clustering is done in same order in both methods but the distances used are different (Ex: 11 and 5.25 while clustering (1,2) and (3,4) together).

[Q2 Continued...]

(b) Obtained $P = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$

For variance of first principle component based on P ,

we have to find $\max x^T P x$ such that $x^T x = 1$.

$f(x) = x^T P x \Rightarrow$ has to be maximized.

Lagrange multipliers method:

$$f(x) = x^T P x + k(1 - x^T x)$$

$\hookrightarrow = 0$

$$f'(x) = 2Px + k(-2x) = 0$$

($\because$ for symmetric matrices,
 $\frac{\partial}{\partial x}(x^T A x) = 2Ax$.)

$$Px = kx$$

k is an Eigen value and x is an eigen vector.

$$x^T P x = x^T (kx) = k(x^T x) = k$$

$$x^T P x = k.$$

Extreme values of $x^T P x$ are eigen values.

Max value of $x^T P x$ is max eigen value, $\lambda_1 \rightarrow$ first principle component
 Variance of first principle component is, $e_1^T P e_1$, where e_1 is
 the eigen vector corresponding to eigen value λ_1 .

for eigen values of P ,

$$|P - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)((1-\lambda)^2 - 1/2) + 1/\sqrt{2}(-1/\sqrt{2}(1-\lambda)) = 0$$

$$(1-\lambda)((1-\lambda)^2 - 1/2) - 1/2(1-\lambda) = 0$$

$$(1-\lambda)((1-\lambda)^2 - 1) = 0$$

$$(1-\lambda)(\lambda^2 - 2\lambda + 1 - 1) = 0$$

$$(\lambda - 1)(\lambda^2 - 2\lambda) = 0$$

$$\lambda(\lambda - 1)(\lambda - 2) = 0$$

eigen values are $\lambda_1 = 2, \lambda_2 = 1, \lambda_3 = 0$

$\lambda_1 = 2$ is the ~~max~~ ~~principal~~ component. is the Max. Eigen value.

for e_1 , $Pe_1 = \lambda_1 e_1$

$$\begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \\ e_{13} \end{bmatrix} = \begin{bmatrix} 2e_{11} \\ 2e_{12} \\ 2e_{13} \end{bmatrix}$$

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$$\Rightarrow \begin{bmatrix} e_{11} + \frac{1}{\sqrt{2}} e_{13} \\ -e_{12} + \frac{1}{\sqrt{2}} e_{13} \\ \frac{1}{\sqrt{2}} e_{11} + \frac{1}{\sqrt{2}} e_{12} + e_{13} \end{bmatrix} = \begin{bmatrix} 2e_{11} \\ 2e_{12} \\ 2e_{13} \end{bmatrix}$$

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∴ we will obtain e_1 from these 3 equations.

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And, Variance of first principle component = $\lambda_1 = 2$.